University Of Engineering and Technology, Lahore Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	**Course Code: CMPE-112L**
Assignment Type: Lab Report	**Dated: 26th March,2026**
Semester: 1st	**Session: 2026**
Lab/Project/Assignment #: 7	**CLOs to be covered: CLO3**
Lab Title: List, Tuple	**Teacher Name: Hafiz Muhammad Mubashar**

Lab Evaluation:

CLO3	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	**Level2**	**Level3**	**Level4**	**Level5**	**Level6**
(5)						
					Total	/5

Rubrics for Current Lab Evaluation

Scale	**Marks**	**Level**	**Rubric**
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 7: List, Tuple

Lab Goals/Objectives:

To understand and use Python's list and tuple data structures, including creation, indexing, slicing, and the key differences between mutable lists and immutable tuples.

Equipment Required:
Python 3, VS Code / PyCharm

Theory:

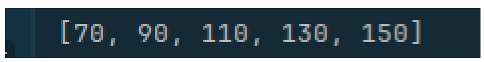
A list is an ordered, mutable collection of items that can be changed after creation (e.g. by adding, removing, or updating elements). A tuple is an ordered, immutable collection, meaning its contents cannot be changed once created. Both support indexing and slicing to access individual elements or subsequences.

Lab Work:

Task 1: Sum list

```
l1 = [10,20,30,40,50]
l2 = [60,70,80,90,100]
sum_list = []
for i in range (len(l1)):
   sum = l1[i] + l2[i]
   sum_list.append(sum)
print(sum_list)
```

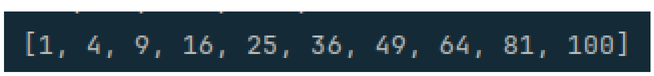
Result:



Task 2: Multiplied List

```
l1 = [1,2,3,4,5,6,7,8,9,10]
mul_list = []
for e in l1:
   mul_list.append(e**2)
print(mul_list)
```

Result:



Task 3: Even and Odd List

```
l1 = [0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20]
even_list = []
odd_list = []
for e in l1:
   if e % 2 == 0:
      even_list.append(e)
   else:
      odd_list.append(e)
print(f"even_list: {even_list}")
```

print(f"odd_list: {odd_list}")

Result:

```
even_list: [0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
odd_list: [1, 3, 5, 7, 9, 11, 13, 15, 17, 19]
```

Task 4: Creating and Accessing a List
fruits = ["apple","banana","mango","grape"]
print(fruits)
print(fruits[0])
print(fruits[-1])
Result:

```
['apple', 'banana', 'mango', 'grape']
apple
grape
```

Task 5: List Slicing
numbers = [10,20,30,40,50]
print(numbers[1:4])
print(numbers[:3])
print(numbers[2:])
Result:

```
[20, 30, 40]
[10, 20, 30]
[30, 40, 50]
```

Task 6: Modifying a List
marks = [80,90,70]
marks[1] = 95
marks.append(100)
print(marks)
marks.remove(70)
print(marks)
Result:

```
[80, 95, 70, 100]
[80, 95, 100]
```

Task 7: Looping through a Tuple

```
subjects = ("Math","Physics","Programming")
for i in range(len(subjects)):
    print(subjects[i])
```

Result:

```
Math
Physics
Programming
```

Task 8: Tuple Immutability

```
student = ("Anaya",19,"CE-CYS")
print(student)
print(student[1])
# student[1] = 20   tuples cannot be changed so it will not mutate
```

Result:

```
('Anaya', 19, 'CE-CYS')
19
```

Conclusions:

Lists and tuples were used to store and manage collections of data. List operations such as indexing, slicing, and modification (append, remove) were practiced, while tuples were used to demonstrate immutability and unpacking, highlighting the key difference between mutable and immutable sequences in Python.